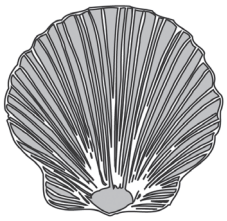
Limestone

☞ **1** Read the passage below and underline five characteristics of limestone.



Limestone is a type of rock that formed millions of years ago from the shells of dead ocean creatures. The shells built up on the sea floor and, over time, changed into rock. The chemical name for limestone is **calcium carbonate**. It has the chemical formula $CaCO_3$. Limestone is dug out of the ground and used for many things, especially as a building material. It is used for making cement, mortar, concrete and building blocks.

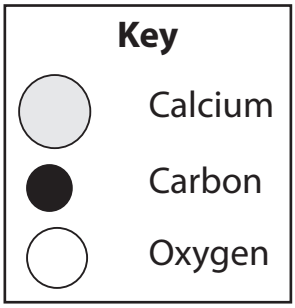
Limestone can be damaged by **acid rain**. When fossil fuels are burnt they release chemical pollutants such as sulfur dioxide and nitrogen dioxide. These chemicals react with water in the atmosphere to form acid rain. Normally rain is slightly acidic, but anything with a pH of less than 5.6 is known as acid rain. Acid rain eats away at limestone, causing it to crumble.

2 In your workbook, write down the five characteristics you underlined in point form.

3 Complete the following sentence.

A molecule of calcium carbonate or $CaCO_3$ is made up of one atom of _____ and _____, and _____ atoms of oxygen.

4 Using the key, draw a molecule of calcium carbonate.



5 Match the sentence halves to complete the statements below.

When fossil fuels are burnt they	with water to form acidic rain.
The chemical pollutants released combine	to form sulfuric acid.
Acidic rain has a pH of less	release sulfur dioxide and nitrogen dioxide.
Sulfur dioxide combines with water	than 5.6.

6 Choose the correct ending for each of the statements below.

The chemical formula for limestone is	$CaCO_{2.}$	$CaCO_{3.}$
The chemical formula for nitrogen dioxide is	$NO_{2.}$	$NO_{3.}$
The chemical formula for sulfuric acid is	$HNO_{3.}$	$H_2SO_{4.}$

© Janine Haeusler and Blake Education 2014
ISBN 978 1 92571 062 5